

This project aimed to rigorously test whether the implementation of a retry payment feature significantly reduces payment dropouts across various payment methods. The primary hypothesis posited that the retry feature would indeed reduce payment dropouts (H_1), contrasting with the null hypothesis (H_0) that it would have no effect. A secondary hypothesis investigated whether the retry success rate differed significantly from the first attempt success rate. After loading the dataset, a crucial step involved calculating the required sample size, which determined a total of 20 participants (10 per group) based on an empirical baseline dropout rate of 0.1479 and a target dropout rate of 0.0978. Data cleaning followed, where duplicate entries were removed, and missing values were handled, resulting in a robust dataset of 9946 rows, evenly distributed between the control (5032 users) and test (4914 users) groups, ensuring each user belonged to a single group. The distribution across payment methods (Debit Card, Netbanking, UPI) was also balanced between groups. Visualizations provided initial insights, illustrating retry usage rates, overall payment success rates (control vs. test), dropout rates by group and payment method, the retry funnel, and a comparison of first versus retry success rates within the test group.

For the primary hypothesis, a one-sided Z-test for proportions was conducted, revealing a control dropout rate of 0.1481 compared to a test dropout rate of 0.0999. The resulting Z-statistic of 7.2752 and a p-value of 0.0000 indicated a statistically significant reduction in payment dropouts attributable to the retry feature, leading to the rejection of the null hypothesis. Furthermore, Cohen's h calculated as 0.1467, suggested a small yet meaningful effect size. An additional Chi-square independence test demonstrated a highly significant association between users clicking the retry option and achieving final payment success ($\chi^2=109.0607$, $p\text{-value} = 0.0000$), affirming the retry feature's positive impact. The secondary hypothesis test, comparing first attempt (0.8523) and retry attempt (0.7298) success rates, yielded a Z-statistic of 5.8774 and a two-sided p-value of 0.0000, confirming a statistically significant difference between these rates. Finally, multiple testing correction (Bonferroni and Holm) was applied to the payment method-specific p-values (UPI, Debit Card, Netbanking), all of which remained statistically significant after adjustment (adjusted $\alpha = 0.0167$), reinforcing that the retry feature consistently reduces dropout across all tested payment methods.

Interpretations and Recommendations:

The analyses provide strong statistical evidence that implementing a retry payment feature significantly reduces payment dropouts, with a small but positive effect size. There is also a clear association between users utilizing the retry option and ultimately achieving payment success. While the success rate for retry attempts is observed to

be lower than that of first attempts, the feature demonstrably contributes to an overall reduction in payment dropouts across all major payment methods.

Based on these findings, the following recommendations are made:

- **Full Implementation:** Proceed with the full implementation of the retry payment feature, as it has proven to be an effective strategy for improving overall payment completion rates.
- **Feature Optimization:** Investigate the reasons behind the lower success rate of retry attempts compared to first attempts. This could involve analyzing error codes for retries, refining the retry logic, or providing clearer guidance to users during the retry process to further enhance its effectiveness.
- **User Experience (UX) Enhancement:** Continuously monitor user interactions with the retry feature and seek opportunities to improve the user experience, making the retry process as seamless and intuitive as possible.
- **Ongoing Monitoring:** Establish key performance indicators (KPIs) related to payment dropout rates and retry success rates, and continuously monitor these metrics to ensure the feature continues to perform optimally and to identify any future areas for improvement.